

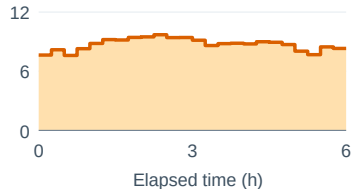
Queued AI work follows renewable surplus

Joint admission, start-time and GPU power-cap scheduling

a Empirical inputs

CAISO renewable curtailment

System-wide power (GW)



20 six-hour events · one global scale

18 production Spot jobs

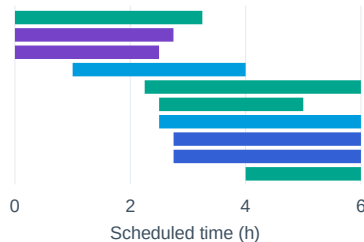
Alibaba A100 trace · 256 GPUs in queue

Field-measured power + throughput

b Joint scheduling

Admit · start · power cap

Event 0: 10 of 18 jobs admitted



250 W 300 W 350 W 400 W

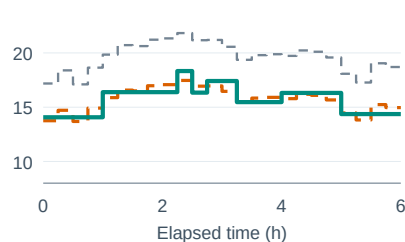
D-Wave hybrid · 945 binary choices

30 s requested solver time

c Dispatch delivery

Event 0 · flexible GPU power

Power (kW)



Scheduled load

80% target

Envelope

Peak flexible envelope: 23.52 kW

20-EVENT CAMPAIGN

Modelled GPU schedules

0.511 kW

Mean tracking RMSE

99.94%

Mean energy / target

210 GPU-h

Mean baseline-equivalent work